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The Noise on Ultraviolet Astronomical Images Captured by Back-illuminated Silicon Sensors

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ABSTRACT

Ultraviolet (UV) astronomy currently relies on back-illuminated silicon imaging sensors. While the noise in these sensors is typically assumed to be dominated by photon shot noise, measurements from HST's Wide Field Camera 3 (WFC3) and the Interface Region Imaging Spectrograph (IRIS) reveal a significant discrepancy: the noise measured in the ultraviolet (UV) is systematically lower than theoretical predictions. We propose that this discrepancy is caused by partial charge collection (PCC), a whereby a fraction of photogenerated electron-hole pairs recombine before they can be measured. We present a simple theoretical model, valid for wavelengths from 1 Å to 10 000 Å, that incorporates the effects of PCC and shows better agreement with the noise measurements from both WFC3 and IRIS, resolving the previously unexplained discrepancy. This finding implies that the signal-to-noise ratio achievable with these sensors is higher in the UV than previously understood, potentially impacting both future instrument proposals and the uncertainties of our current UV imagery. At about 2000 Å to 3500 Å, PCC is the dominant noise source in the high-signal limit.

Keywords: Astronomical instrumentation (799); Astronomical detectors (84); Ultraviolet astronomy (1736)

1. INTRODUCTION

Back-illuminated, silicon-based image sensors such as charge-coupled devices (CCDs) and complementary metal–oxide–semiconductor (CMOS) sensors are ubiquitous in ultraviolet (UV) astronomy, and are currently used in many UV instruments such as the Atmospheric Imaging Assembly (AIA) (?), the Interface Region Imaging Spectrograph (IRIS) (?), and HST's Wide Field Camera 3 (WFC3) (?). Predicting the noise measured by these sensors is important both for quantifying the uncertainty on current astronomical observations as well as for evaluating the performance of proposed future instruments.

In the high-signal limit (that is, neglecting readout noise), the largest source of noise measured by these sensors is thought to be shot noise (?). This noise is due to the quantized nature of light (?) and is not a property of the sensor *per se*, but its magnitude depends on the number of measured photons, which is determined by the sensor's geometry and optical constants. Another well-known source of noise inherent to silicon sensors is Fano noise (?), the random variation of the number of electrons produced per interacting photon. This noise source is small (?), and often ignored entirely, but it can be combined with the shot noise to form the traditional estimate of the total noise (??).

This traditional model of shot noise and Fano noise is very accurate in the visible (?) and X-ray (???) wavelength regimes. However, this model appears to be less accurate in the UV. For example, ? showed that IRIS measured *less* noise in the far ultraviolet (FUV) channel than the traditional model would suggest and similar results were found in the near ultraviolet (NUV) by WFC3 (?). This discrepancy indicates that the traditional model is incomplete, and that a more detailed model is needed to predict the noise measured by these sensors in the UV.

In this work, we aim to resolve the UV noise discrepancy by developing a noise model which adds two additional effects to the traditional model. The first effect is partial charge collection (PCC), which occurs when electron-hole pairs recombine before they can be measured by the sensor (??). Noise due to PCC is a well-known effect, for example ? accounted for PCC in their measurement of the Fano noise in soft X-ray (SXR), but its influence on the noise measured in UV has been rarely modeled in detail. The second effect is charge diffusion (or charge spreading), the tendency for photoelectrons to migrate into neighboring pixels (?), a mechanism suggested by both ? and ? to explain the discrepancies

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they measured. We will model the contribution of each of theses effects to the total noise and compare our predicted noise to the noise measured by IRIS and WFC3. We will also provide noise predictions for AIA and the Multi-slit Solar Explorer (MUSE) (**?**) which can be used for estimating the uncertainty of observations captured by these instruments. Finally, we have made available a reference implementation of our noise model in Python which is available to be installed from the Python Package Index (PyPI).

2. SENSOR MODEL

In this work, we will use the back-illuminated CCD model described in **?** as a basis for our sensor model to make it directly comparable with previous works in the literature. Figure 1 is a schematic of our sensor model and it shows the light-sensitive region of the sensor to be an epitaxial silicon layer with a thickness D. A fraction of the incident photons are either absorbed in the oxide layer (thickness δ) or reflected at the first or second interface. The illuminated side of the epitaxial layer is considered to have a PCC region of thickness W, where electron-hole pairs can recombine before being measured, and there is a depletion region of thickness z_d on the non-illuminated side.

In Section **??** we will calculate the signal predicted by this sensor model using the method described in **?**, adapted to more recent measurements. In Section **??** we will calculate the noise predicted by this sensor model in a self-consistent manner, and in Section **??** we will model the charge diffusion of this sensor which is needed for a complete description of the noise. Throughout this section we will adopt the convention that unprimed variables represent quantities incident on the sensor, primed variables represent quantities internal to the sensor, and the double-primed variables represent quantities measured by the sensor.

2.1. *Signal*

The average number of photoelectrons measured per incident photon is known as the quantum efficiency (QE). This is a common performance metric for measuring sensor sensitivity and it is given in **?** as

$$\mathrm{QE}(\lambda, T) = \left\langle \frac{N_e''}{N_\gamma} \right\rangle = A(\lambda)\,\overline{n}(\lambda, T)\,\overline{\eta}(\lambda), \qquad (1)$$

where λ is the wavelength of the incident photons in vacuum, T is the temperature of the sensor, $\langle X \rangle = \overline{X}$ denotes the expected value of a random variable X, N_e'' is the number of electrons measured by the sensor, N_γ is the total number of photons incident on the sensor, $A(\lambda)$ is the absorbance of the light-sensitive layer, $\overline{n}(\lambda, T)$ is

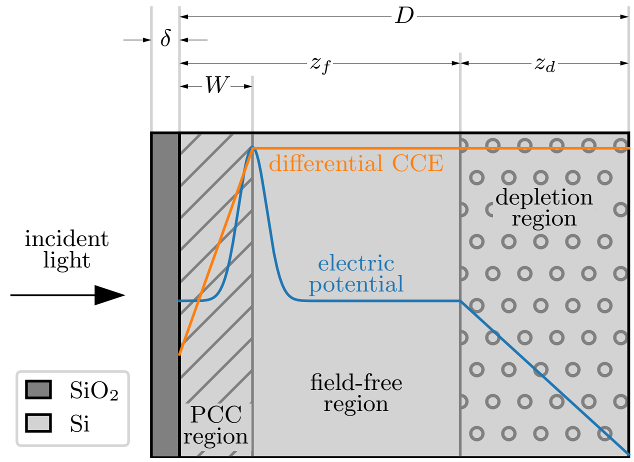

Figure 1. A schematic (not to scale) of the back-illuminated sensor model used in this work, labeled with the thicknesses of each layer. Overplotted is a *qualitative* example of the electric potential within the sensor that motivates the differential CCE.

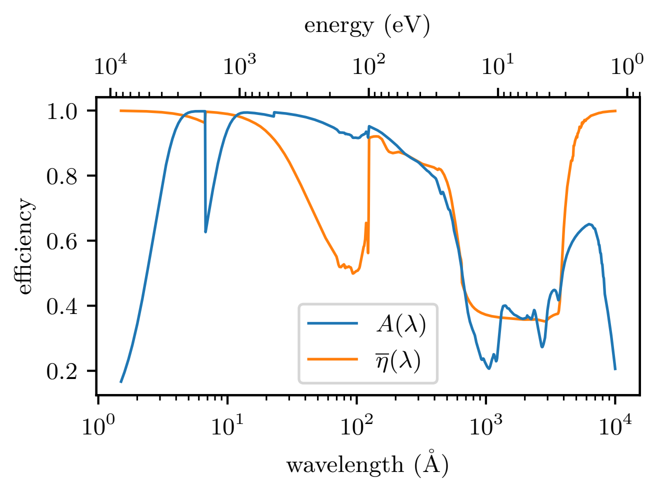

Figure 2. The fraction of incident light absorbed by the light-sensitive silicon layer and the CCE as a function of wavelength for the parameters inferred from the **?** measurements.

the average quantum yield, the number of photoelectrons generated per absorbed photon, and $\overline{\eta}(\lambda)$ is the average charge-collection efficiency (CCE), the fraction of generated photoelectrons measured by the sensor.

The fraction of incident energy that is absorbed by the light sensitive layer is the absorbance, $A(\lambda)$, which is reduced by reflections at each interface, absorption in the oxide layer (UV), or by penetration through the device (SXR and infrared (IR)). $A(\lambda)$ can be determined from the optical constants of Si and SiO_2 using, for example, the popular IMD code (**?**). For this work, we used our Python library, `optika` (**?**), which uses the transfer ma-

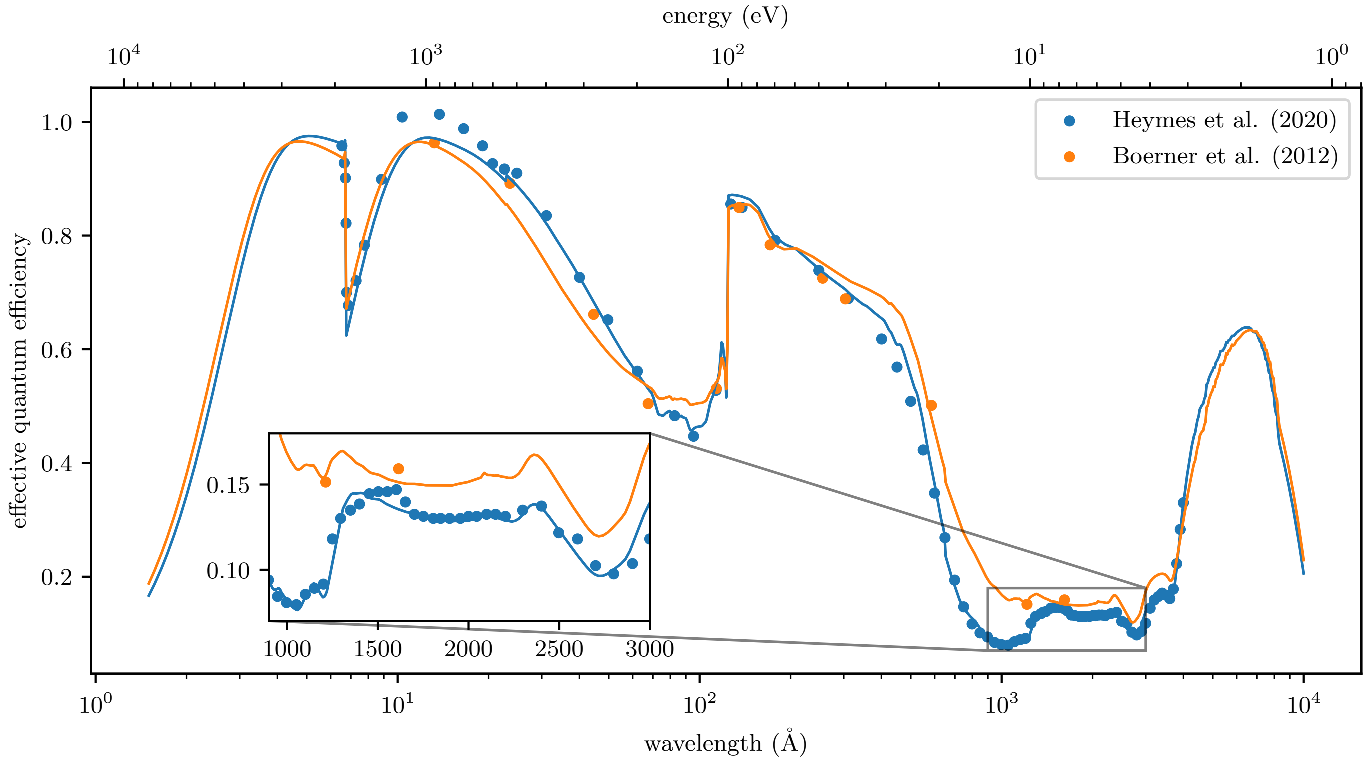

Figure 3. A comparison of EQE(λ) measured by **?** and **?**, along with the best-fit models described in Table **??** plotted as lines with the same color as the data. The inset zooms into the ultraviolet to better visualize the difference between the measurement and model in this range.

Table 1. The sensor model parameters which best fit the measurements in **?** and **?**.

Source	implementation	η_0	W (nm)	δ (nm)	D (μm)	substrate roughness (nm)
?	`e2v_ccd203`	0.452	335	1.4	16	1.1
?	`e2v_ccd97`	0.334	153	4.6	14	4.8

trix method described in **?** with the optical constants from **?**, **?**, and **?** to compute the electric field for every interface in the sensor. In Figure **??** we have plotted $A(\lambda)$ for the **?** parameters in Table **??**.

In **?**, the authors assume no reflections from the unilluminated side of the sensor for simplicity. In this work, we compute the total change in Poynting flux into and out of the light-sensitive region of the sensor to determine $A(\lambda)$. This treatment introduces interference effects for infrared wavelengths, which can be seen on the right side of Figure **??**.

Determining $\overline{n}(\lambda, T)$ over the entire wavelength range considered in this study is an area of ongoing research (**????**, just to name a few). This work will use the quantum yield model developed by **?** which uses a phenomenological model of impact ionization to bridge the "UV gap", an area where direct measurements of the quantum yield are not yet available. They give the average quantum yield as

$$\overline{n}(E,T) = \begin{cases} \sum_{n=0}^{\infty} n p_n(E,T), & 0 < E \leq 50\,\mathrm{eV} \\ E/\epsilon_{eh}(T), & 50\,\mathrm{eV} < E < \infty, \end{cases} \tag{2}$$

where E is the energy of the incident photon, n is the actual quantum yield, $p_n(E,T)$ is a table of pair-creation probabilities which is provided in their supplementary material,

$$\epsilon_{eh}(T) = 1.7 E_g(T) + (0.084\,\mathrm{eV}^{-1}) A + 1.3\,\mathrm{eV}, \tag{3}$$

is the asymptotic mean energy per electron-hole pair, $A = 5.2\,\mathrm{eV}^2$, and

$$E_g(T) = 1.1692\,\mathrm{eV} - \frac{(4.9 \times 10^{-4}\,\mathrm{eV\,K^{-1}})\,T^2}{T + 655\,\mathrm{K}}, \tag{4}$$

is the bandgap energy of silicon.

In **?**, the average CCE is expressed in terms of a differential CCE, $\eta(z)$, which is the fraction of photoelectrons collected for a photon absorbed at a depth z into the epitaxial layer. The average CCE is then the first moment of the differential CCE weighted by the probability of absorbing a photon at a depth z,

$$\overline{\eta}(\lambda) = \frac{\int_0^\infty \eta(z)\, e^{-\alpha(\lambda) z}\, dz}{\int_0^\infty e^{-\alpha(\lambda) z}\, dz}, \tag{5}$$

where $\alpha(\lambda)$ is the absorption coefficient of silicon for the given wavelength.

In principle, $\eta(z)$ is a function of the exact implant profile which is usually impractical to measure, but see **??** for a case where the authors did have a measurement of the implant profile provided by the manufacturer. In **?**, the authors instead adopt a piecewise-linear approximation of the differential CCE,

$$\eta(z) = \begin{cases} \eta_0 + (1 - \eta_0) z / W, & 0 < z < W \\ 1, & W < z < \infty \end{cases} \tag{6}$$

where η_0 is the differential CCE at the back surface of the sensor. Plugging Equation **??** into Equation **??** yields an expression for the CCE,

$$\overline{\eta}(\lambda) = \eta_0 + \left(\frac{1 - \eta_0}{\alpha W} \right) (1 - e^{-\alpha W}), \tag{7}$$

which can be used in Equation **??** to determine the QE. In Figure **??** we have plotted the average CCE for the **?** parameters in Table **??**.

In **?**, the authors define an effective QE as

$$\text{EQE}(\lambda) = A(\lambda)\, \overline{\eta}(\lambda), \tag{8}$$

which is found by taking the ratio of the current measured by the sensor to the current measured by a calibrated photodiode and is the quantity that is typically measured when calibrating a image sensor (**???**). In Figure **??**, we have plotted the measured, effective QE for two sources: **?** which measured the AIA CCDs at a few discrete wavelengths, and **?** which measured a Teledyne e2v CCD97 sensor over a wide wavelength range with high resolution using a monochromator.

Using the Nelder-Mead minimization algorithm (**?**), we found the free parameters of our model which best fit the data in **?** and **?**. These models are plotted as solid lines in Figure **??**, and the corresponding values of the free parameters are shown in Table **??**. Throughout the remainder of this work we will use the model which best fits the **?** data.

2.2. *Noise*

Our noise model will consider three sources: shot noise from the quantized nature of the photons striking the sensor, Fano noise due to inherent randomness in the process which converts photons to electrons, and noise due to electrons randomly recombining before they can be measured by the sensor. This section will describe the statistics of each noise source and demonstrate a procedure which can simulate the noise measured by our model sensor for a given number of incident photons.

Throughout this work, we will measure noise in terms of a variance-to-mean ratio (VMR),

$$F(X) = \frac{\text{Var}(X)}{\overline{X}}, \tag{9}$$

where $\text{Var}(X)$ is the variance of X. Using the VMR instead of the signal-to-noise ratio (SNR) to express the noise is convenient since it is constant as a function of signal for the distributions studied here. For example, the VMR of a Poisson random variable is always unity since its variance and expectation value are equal. The VMR is not dimensionless (it has the same units as X), so we must take care to interpret the VMR in terms of the correct units. For a constant multiple of a random variable, aX,

$$F(aX) = aF(X). \tag{10}$$

We will use this relationship throughout this work to convert from incident photons to measured electrons and vice versa.

In Figure **??** we have plotted the VMR for the noise sources considered in this study in two different units: number of incident photons and number of measured electrons. Figure **??**a is useful when *designing* an instrument since the number of photons incident on the sensor can be determined by the radiance of the source and the effective area of the instrument. Figure **??**b is useful when *calibrating* an instrument since we directly measure these electrons. The details of the calculations for Figure **??** are presented in the subsections below.

2.2.1. *Shot Noise*

Photon shot noise is often assumed to be the leading noise contributor in UV astronomy (**???**). If the expected number of photons that interact with the light-sensitive layer is

$$\overline{N'_\gamma} = A \overline{N_\gamma}. \tag{11}$$

then the actual number of interacting photons, N'_γ, is sampled from the Poisson distribution,

$$N'_\gamma \leftarrow \text{Poisson}\big(\overline{N'_\gamma}\big). \tag{12}$$

As mentioned in the previous section, the VMR of this process is

$$F(N'_\gamma) = 1, \tag{13}$$

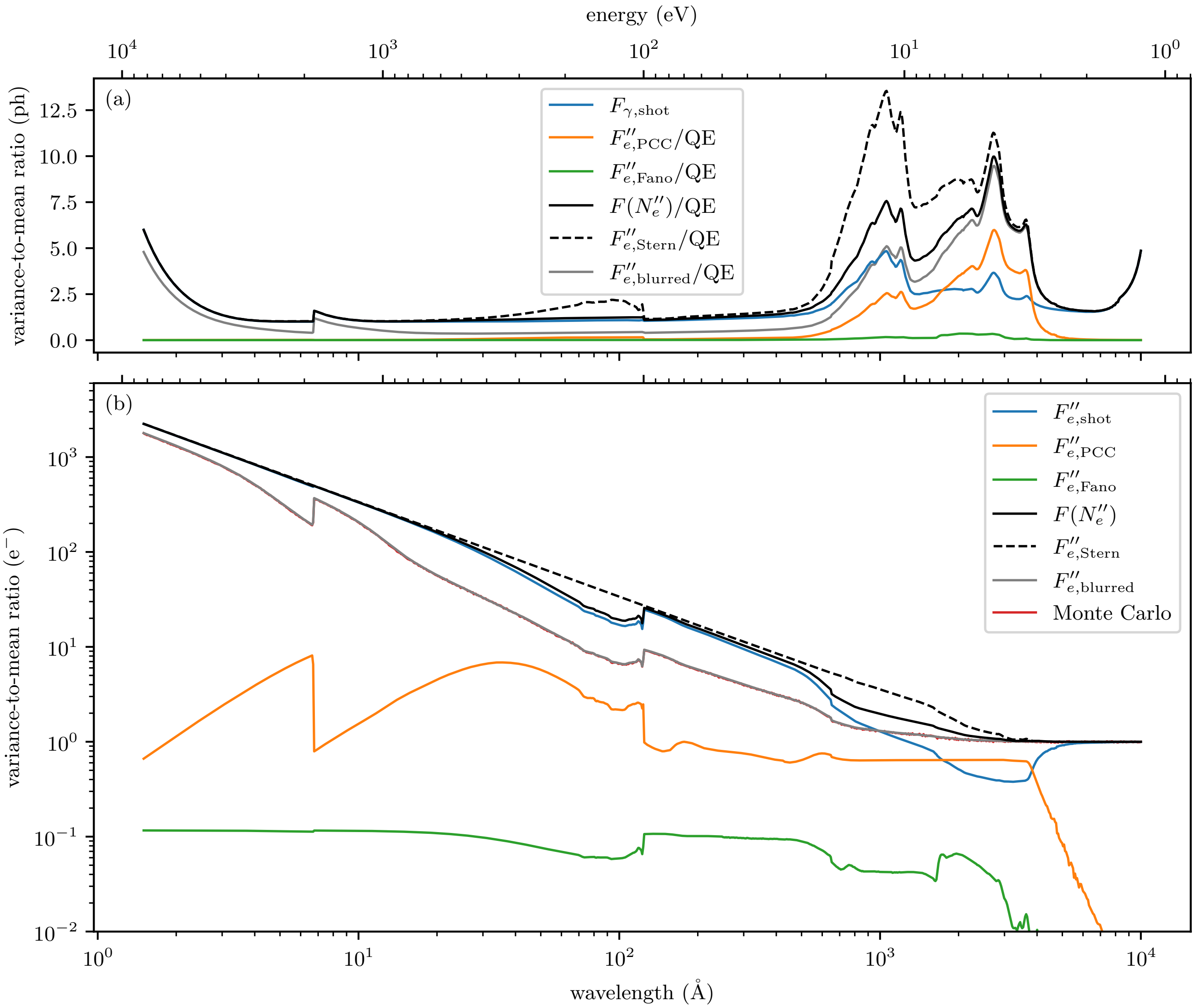

Figure 4. The total and component-wise VMR of our sensor model expressed in two different units: photons incident on the sensor (top) and electrons measured by the sensor (bottom). Plotted for comparison is the VMR of the **?** noise model (dashed) and a Monte Carlo simulation of $F(N''_e)$ in red.

but this in terms of the number of absorbed photons, which is internal to the sensor. In terms of the expected number of incident photons given the number of absorbed photons the VMR is

$$F_{\gamma,\text{shot}} \equiv F\left(\frac{N'_\gamma}{A}\right) = \frac{1}{A} \tag{14}$$

since A is the conversion factor between incident photons and absorbed photons. $F_{\gamma,\text{shot}}$ is plotted in Figure **??**a in blue and demonstrates that this expression is nearly unity across almost the entire wavelength range of the sensor except in the UV, where the absorbance is poor (Figure **??**). We can also express the VMR of the shot noise in terms of the number of measured electrons by using the QE as the conversion factor between the unprimed and double-primed variables,

$$F''_{e,\text{shot}} = \overline{n}\,\overline{\eta}. \tag{15}$$

This equation is plotted in Figure **??**b in blue, where we can see that it increases linearly with decreasing wavelength since the average quantum yield is increasing.

2.2.2. *Fano Noise*

The energy resolution of silicon detectors is ultimately limited due to Fano noise (**?**), the random variation of the quantum yield n. Fano noise is usually expressed in terms of the Fano factor, $\mathcal{F} = F(n)$. Silicon is commonly accepted to have $\mathcal{F} \approx 0.1$ (**?**). In part due to variations of the Fano noise as a function of wavelength and temperature (**?**), there is some disagreement in the literature around a more precise value for $\mathcal{F}$ (**??????**, & references therein). However, because the Fano noise is so small compared to the other noise sources considered in this study, the precise form of this noise model does not appreciably influence our conclusions.

As we mentioned in Section **??**, this work uses the quantum yield model of **?**. They provide the probability mass function (PMF) of the quantum yield distribution, $p_n(E,T)$, which we can use directly to generate random samples of the quantum yield for $E \leq 50\,\mathrm{eV}$. For $E > 50\,\mathrm{eV}$, we will assume that the quantum yield is a Gaussian distribution centered around $\overline{n}(\lambda,T)$, with standard deviation $\sqrt{\overline{n}(\lambda,T)\,\mathcal{F}(T)}$, where $\mathcal{F}(T)$ is the asymptotic Fano factor given by **?** as

$$\mathcal{F}(T) = (-0.028\,\mathrm{eV}^{-1})\,E_g(T) + (0.0015\,\mathrm{eV}^{-2})\,A + 0.14. \tag{16}$$

At 300 K, this corresponds to a Fano factor of approximately 0.114, which is generally consistent with the best-available measurement of the Fano factor by **?** at 6 keV.

$\mathcal{F}$ is a quantity internal to the sensor. The measured Fano factor is instead

$$F''_{e,\mathrm{Fano}} = \overline{\eta}\mathcal{F} \tag{17}$$

since the conversion factor between the generated electrons and the measured electrons is the CCE. In Figure **??**b we have plotted $F''_{e,\mathrm{Fano}}$ in green. This shows that the Fano noise is the smallest contributor to the total noise across the entire wavelength range and is often at least an order of magnitude smaller than the other noise sources.

2.2.3. *Partial Charge Collection Noise*

In Figure **??**, we have plotted the penetration depth in silicon vs. the thickness of the PCC region. We can see that there are two regions in the UV where the penetration depth is less than the thickness of the PCC region. In these regions, there is a significant chance that some of the electron-hole pairs will randomly recombine before being measured by the sensor. The stochastic nature of this process leads us to consider this as a new noise source, PCC noise.

In the **?** charge-collection model, each photoelectron generated in the PCC region has a probability $\eta(z)$ of *not* recombining and subsequently being measured by the sensor. If n'_i is the quantum yield of the ith photon generated using the **?** model discussed in the previous section, we can express the *measured* quantum yield as

$$n''_i \leftarrow \mathrm{Binomial}\big(n'_i, \eta(z_i)\big), \tag{18}$$

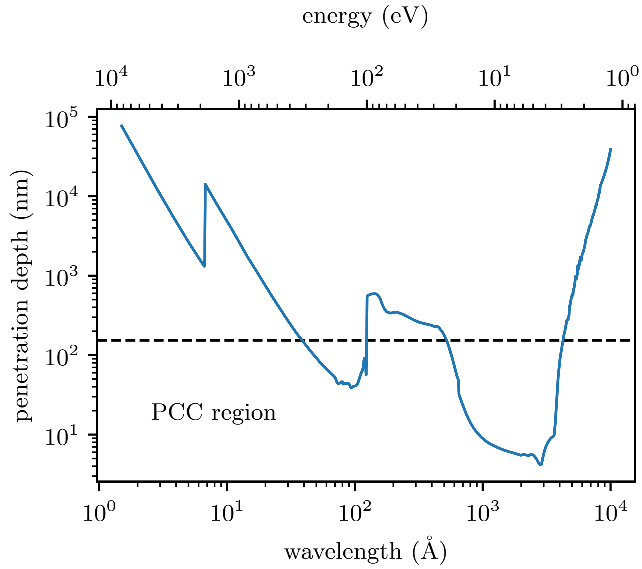

Figure 5. The penetration depth in silicon as a function of wavelength plotted against the depth of the PCC region.

where z_i is the penetration depth of the ith photon and $\mathrm{Binomial}(n,p)$ is a sample from the binomial distribution with n trials and probability of success p. We can use **??** to compute the total number of measured electrons given N'_γ absorbed photons as

$$N''_e = \sum_{i=0}^{N'_\gamma} n''_i. \tag{19}$$

This expression is intended to be used in an instrument forward model to sample the distribution of measured electrons given an expected number of incident photons.

The VMR of this process is non-trivial since Equation **??** is a sum of Binomial draws with completely independent n and p for each term. However, if we compute the VMR of the CCE by taking the second moment of Equation **??** in a similar fashion to Equation **??**,

$$F(\eta) = \frac{2e^{-\alpha W}}{\overline{\eta}} \left(\frac{1-\eta_0}{\alpha W}\right)^2 \big(\sinh(\alpha W) - \alpha W\big), \tag{20}$$

we can use the expressions given by **?** to find the VMR of only the Fano noise and PCC noise as

$$F''_{e,\mathrm{sensor}} = 1 + (\mathcal{F}-1)\overline{\eta} + \big(\overline{n} + \mathcal{F} - 1\big)F(\eta). \tag{21}$$

The contribution from only PCC noise is then

$$F''_{e,\mathrm{PCC}} = F''_{e,\mathrm{sensor}} - F''_{e,\mathrm{Fano}}, \tag{22}$$

which is plotted in Figure **??** in orange. We can see that the PCC noise is usually very small compared to the shot noise, but in the UV from about 2000 Å to 3500 Å,

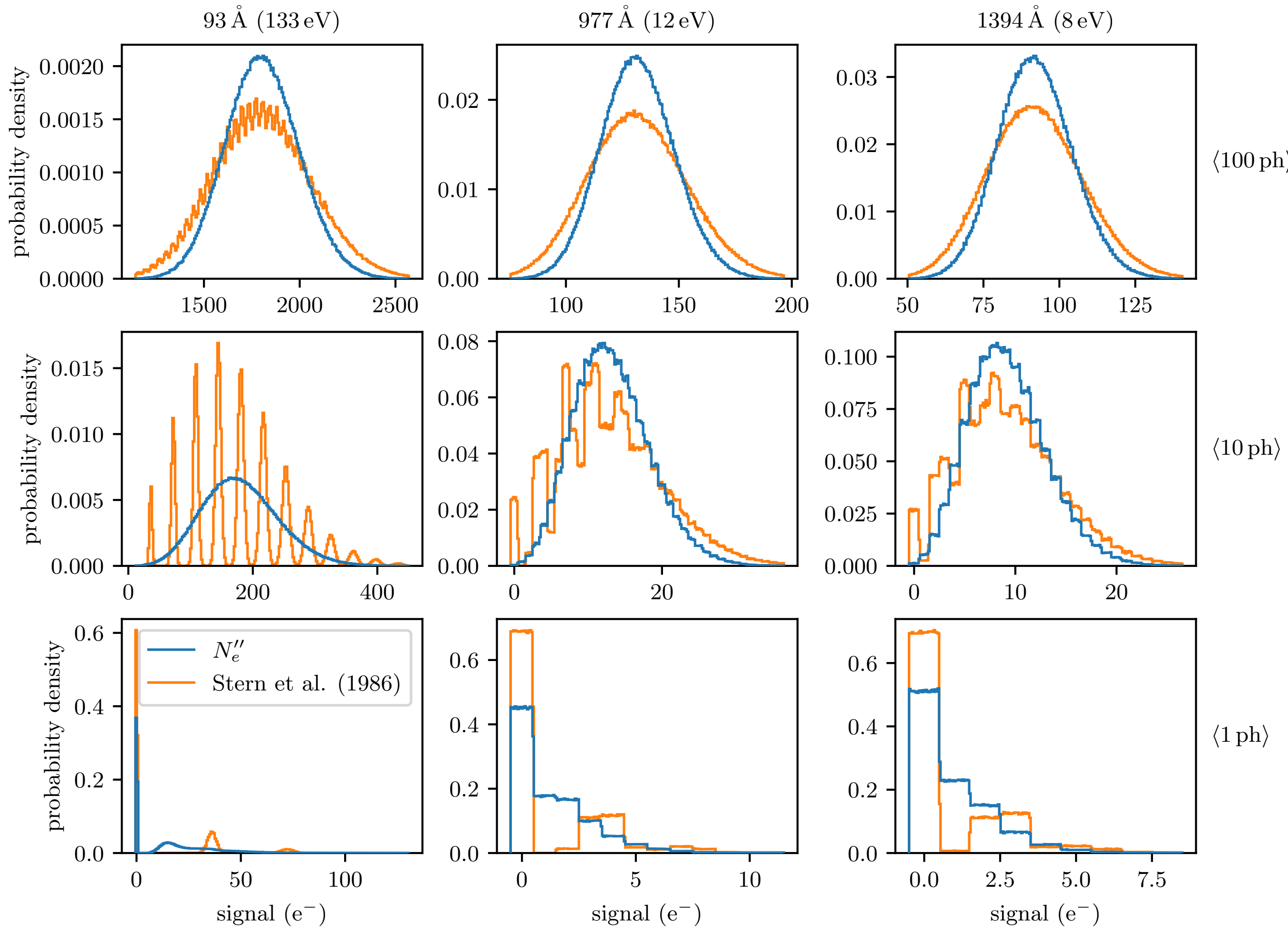

Figure 6. The probability distribution of the number of measured electrons for a given wavelength and expected number of *absorbed* photons calculated using 1000000 samples of Equation **??** and the **?** noise model.

it actually exceeds the shot noise and is the dominant contributor to the noise.

In Figure **??**, we have plotted the PMFs of Equation **??** and the **?** noise model to visualize the differences between these two distributions. Each model has been evaluated on a grid where each column of the grid represents a different wavelength (chosen to demonstrate the worst-case differences between the two models) and each row represents a different expected number of absorbed photons. When the number of absorbed photons is low, like in the bottom two rows of Figure **??**, we can see that the **?** model (orange) has a comb-like appearance caused by the Fano noise slightly blurring the PMF of the photon shot noise. In contrast, our model show much less of this comb pattern since the PCC noise tends to blur the distribution further into a single peak. As the number of photons increases, both distributions tend towards a Gaussian, but our model remains narrower.

To ease adoption of this model, we've provided a reference implementation of Equation **??** in Python, `optika.sensors.signal()`, which is designed to be simple to use for existing and future instrument pipelines.

2.3. *Charge Diffusion*

In most back-illuminated imaging sensors used for UV astronomy, the depletion region (the region with significant electric field) does not extend all the way from the gate structure through the device. As a result, there is a so-called field-free region near the back of the sensor where photoelectrons must undergo a random walk to find their way to the depletion region where they can then be conducted to the terminals and measured (**?**). This random walk generally leads to a loss of spatial resolution measured by the sensor since electrons can diffuse to adjacent pixels. It also leads to a reduction in the variance of a flat field measured by the sensor since the blurring due to this diffusion induces a correlation between neighboring pixels.

Using Monte Carlo modeling, **?** found the following expression for the standard deviation of the charge diffusion kernel:

$$\sigma(z) = \begin{cases} z_f \sqrt{1 - z/z_f}, & 0 < z < z_f \\ 0, & z_f < z < D, \end{cases} \tag{23}$$

where z is the distance from the back surface at which the photon is absorbed, $z_f = D - z_d$ is the thickness of the field-free region of the sensor, and z_d is the thickness of the depletion region. Using Equation **??**, we can find the average variance of the charge diffusion kernel by taking an mean across the entire thickness of the sensor weighted by the probability of a photon being absorbed at that depth,

$$\overline{\sigma}^2 = \frac{z_f \left(\alpha z_f + e^{-\alpha z_f} - 1 \right)}{\alpha \left(1 - e^{-\alpha D} \right)}. \tag{24}$$

The thickness of the depletion region or the field-free region is difficult to measure, and depends on the voltage applied to the sensor and the charge collected at the terminals (**?**).

However, **?** did estimate the size of the charge diffusion kernel, for two discrete wavelengths, of a 15 μm-thick (100 Ω-cm resistivity) CCD for the GOES Soft X-ray Imager. We can use these measurements to estimate the size of the depletion region and model the size of the charge diffusion kernel as a function of wavelength. **?** did not directly measure the size of the charge diffusion kernel, instead they measured a quantity they named the mean charge capture (MCC), the fraction of charge captured by the central pixel. Naively, the MCC would be the integral of the charge diffusion kernel over the extent of a pixel. However, since a photon can strike anywhere within the central pixel, we need to convolve with a rectangle function the width of a pixel before integrating. If we assume that the charge diffusion kernel is a Gaussian with variance $\overline{\sigma}^2$, then the MCC is

$$m = \left[\frac{1}{\sqrt{\pi a}} \left(e^{-a} - 1 \right) + \operatorname{erf}\left(\sqrt{a} \right) \right]^2, \tag{25}$$

where $a = d^2 / 2\overline{\sigma}^2$, d is the width of a pixel, and $\operatorname{erf}(x)$ is the error function.

In the top panel of Figure **??**, we have plotted a fit of Equation **??** to the measurements in **?** which found $z_d = 8.7$ μm best matched the data. In the lower panel of Figure **??**, we have plotted the corresponding standard deviation of the charge diffusion kernel as a function of wavelength which predicts that the charge diffusion is reasonably constant over much of the SXR and ultraviolet wavelengths since the penetration depth is low in this regime.

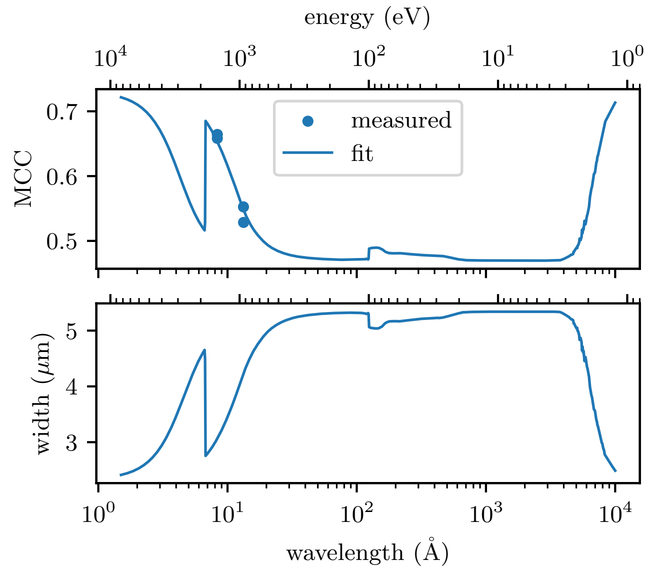

Figure 7. The top panel plots the MCC measured by **?** and the fit of our model. The bottom panel shows the corresponding standard deviation of the charge diffusion kernel for the **?** sensor.

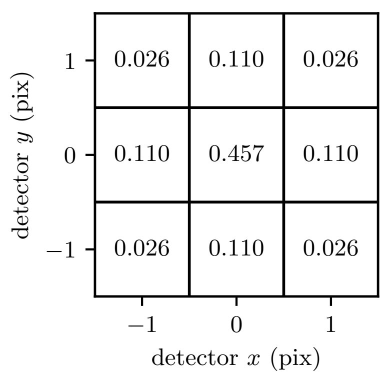

Figure 8. The charge diffusion kernel at 1400 Å convolved with a 13 μm IRIS pixel and integrated over the extent of each pixel.

To forward model this charge diffusion in a practical way, we've included Figure **??**, a 3×3 kernel where the value in each pixel has been computed in a manner similar to Equation **??**, just with different limits of integration. We used the IRIS pixel size and a representative wavelength to demonstrate the worst-case scenario for IRIS. This kernel is intended to be convolved with an input image after the noise model (described above) has been applied to approximate the effects of charge diffusion and complete the forward model of the sensor. We have provided the function `optika.sensors.kernel_diffusion()`, which can be used to compute the charge diffusion kernel for other wavelengths and pixel sizes.

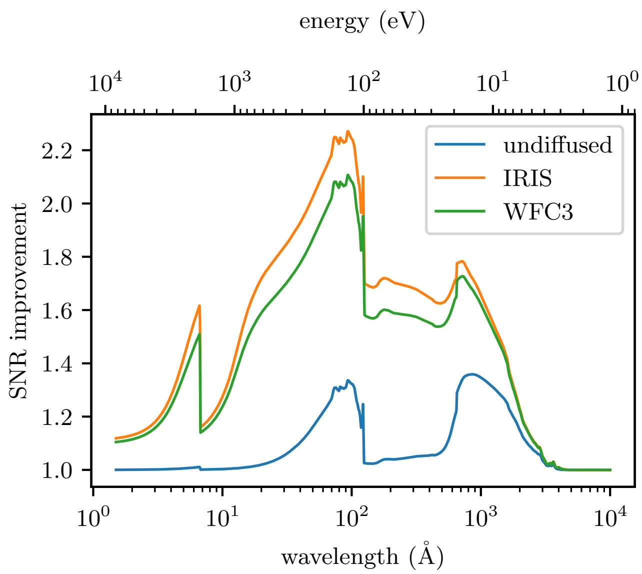

Figure 9. The SNR improvement predicted by our noise model compared to the **?** model. In blue we have plotted the SNR improvement ignoring charge diffusion. In orange and green we have plotted the SNR improvement, including charge diffusion, for a flat-field image observed by IRIS and WFC3.

We can use the definition of a discrete, 2D convolution and Bienaymés formula to find that for an unblurred flat-field image with a given VMR, F_0, the blurring introduced by an even, separable, 3×3 kernel is

$$F_{\text{blurred}} = \epsilon F_0, \tag{26}$$

where

$$\epsilon = \frac{1}{4}\left(3m - 2\sqrt{m} + 1\right)^2. \tag{27}$$

Unlike the other noise sources discussed in Section **??**, the VMR of the charge diffusion process is multiplicative, instead of additive, and cannot be directly compared to these other sources.

3. RESULTS AND DISCUSSION

Since EQE(λ) can be interpreted as an efficiency, it is tempting to think that the expected number of measured photons and their variance is $\text{EQE} \times \overline{N}_\gamma$. However, using the effective QE in this way is equivalent to an all-or-nothing charge collection model where either all the electrons associated with an absorbed photon are measured or none of them are. This simple noise model is formalized in Equations 9 and 10 of **?**, which gives the VMR of the number of measured electrons as

$$F''_{e,\text{Stern}} = \overline{n} + \mathcal{F} \tag{28}$$

This is the noise model used by **?** and **?**. For comparison, the VMR predicted by our undiffused noise model is the sum of $F''_{e,\text{shot}}$ and $F''_{e,\text{sensor}}$,

$$F(N''_e) = 1 + \left(\overline{n} + \mathcal{F} - 1\right)\left(\overline{\eta} + F(\eta)\right). \tag{29}$$

In Figure **??**, we have compared $F''_{e,\text{Stern}}$ (dashed) to our undiffused model, $F(N''_e)$ (black). Over most of the SXR and visible wavelengths the **?** model is a good approximation of the undiffused model developed in this work since the penetration depth is deeper than the PCC region. However, in the UV, where the penetration depth is very shallow, **?** overestimates the noise compared to our undiffused model.

It may seem paradoxical to add a noise source and then measure less total noise, but this is because more photons are measured in our model than the all-or-nothing model. Even though PCC introduces noise, it is less noise than there would be if the photon was undetected. This is good news for designers of UV astronomical instruments since there is less noise than expected from the **?** model in this wavelength range.

In Figure **??**, we have plotted the ratio of the SNR predicted by our model to the SNR predicted by the traditional model. The blue line is the improvement ratio for the undiffused model and it shows that there are two regions, one from 30-100 Å and another from 500-2000 Å, where

where the noise statistics deviate from the traditional model, in some cases by up to ∼30%. This figure can be used to quickly estimate the importance of PCC noise in a given wavelength range.

In Table **??**, we have calculated the VMR in terms of incident photons and measured electrons for the target wavelengths of a few popular and upcoming solar instruments: AIA (**?**), IRIS (**?**), and MUSE (**?**). The results show that for the extreme ultraviolet (EUV) channels, AIA and MUSE are nearly shot-noise-limited since the VMR in units of incident photons is near unity. Tabulated in the second-to-last column of Table **??** is the improvement factor between the VMR of the **?** noise model and our noise model. These ratios show that the AIA 94 Å and 1600 Å, IRIS 1330 Å and 1400 Å, and MUSE 108 Å channels are predicted to have at least 20% more SNR than the traditional noise model would suggest. These results do not include the influence of charge diffusion, so we've included the size of the charge diffusion kernel in the last column which can be used in a forward model of these instrument to blur the result after the noise has been applied.

In Table **??**, we've attempted to reproduce the measurements of **?** and **?** by taking the ratio of the VMR of a simulated UV flat-field image to the VMR of a simulated visible flat-field image. The flat-field images were created by drawing samples from Equation **??**, and then

Table 2. The VMR predicted by our model (including charge diffusion) for prominent wavelengths in selected UV observatories, in both incident photon and measured electron units. Also included is the SNR improvement ratio between our model and the traditional noise model as well as the standard deviation of the charge diffusion kernel.

Instrument	wavelength (Å)	VMR (ph)	VMR (e$^-$)	SNR improvement	diffusion width(μm)
AIA	94	0.43	6.96	2.27	5.32
	131	0.40	8.98	1.69	5.05
	171	0.42	6.72	1.72	5.15
	193	0.44	5.96	1.71	5.17
	211	0.45	5.53	1.70	5.17
	304	0.51	3.96	1.68	5.21
	335	0.53	3.64	1.67	5.21
	1600	3.83	1.15	1.42	5.34
	1700	4.51	1.12	1.35	5.34
IRIS	1330	3.22	1.20	1.52	5.34
	1400	3.23	1.19	1.49	5.34
	2796	9.21	1.02	1.08	5.34
	2832	8.93	1.02	1.09	5.34
MUSE	108	0.43	6.75	2.15	5.32
	171	0.42	6.72	1.72	5.15
	284	0.49	4.21	1.69	5.20
WFC3	2080	6.08	1.07	1.19	5.34
	2240	6.57	1.05	1.15	5.34
	2400	6.40	1.04	1.13	5.34
	2560	7.90	1.03	1.10	5.34
	2720	9.51	1.03	1.09	5.34
	2880	8.51	1.03	1.08	5.34
	3040	6.56	1.02	1.05	5.34
	4000	3.01	1.01	1.01	5.30

Table 3. The ratio of the VMR of a UV flat-field image to the VMR of visible light flat-field image. The second column from the right is the value of this ratio measured by **?** and **?**. The rightmost column is the value of this quantity predicted by our noise model (including charge diffusion).

Instrument	$\lambda_{\rm UV}$ (Å)	$\lambda_{\rm vis}$ (Å)	measurement	model
IRIS	1350	4500	1.50	1.19
WFC3	2080	4000	1.09	1.06
	2240	4000	1.09	1.05
	2400	4000	1.07	1.04
	2560	4000	1.04	1.03
	2720	4000	1.04	1.02
	2880	4000	1.03	1.02
	3040	4000	1.01	1.01

convolving with the appropriate charge-diffusion kernel, such as Figure **??** for IRIS. This table shows that the discrepancy discussed in the introduction is mostly resolved. For IRIS, **?** measured 1.5 at 1350 Å expecting about 2, but our model predicted 1.19, which is much closer. Similarly for WFC3, **?** measured 1.09 at 208 nm expecting about 1.7 and our model predicted 1.06, which again is closer than their expected value. The reason for the remaining discrepancy is not well-understood, but one obvious possibility is that the QE of the sensors on WFC3 and IRIS is slightly different than that of the sensor measured by **?**. Another possibility is that the thickness of the depletion region is different for these sensors than the one measured by **?**, leading to more charge diffusion.

4. CONCLUSIONS AND FUTURE WORK

This work aims to resolve the discrepancy identified by **?** and **?** in the predicted vs. observed noise measured by back-illuminated silicon imaging sensors. To accomplish this, we reconsidered the effects of PCC and charge diffusion on the noise by developing a simple model of a back-illuminated sensor based on the CCE introduced by **?** and a charge diffusion model found by **?**. By analyzing these effects in a self-consistent manner, we were able to mostly resolve the discrepancy and propose that PCC is the most important effect to consider since we predict that the charge diffusion in visible and UV is nearly identical.

Our sensor model predicts that there are two UV bands, 30-100 Å and 500-2000 Å, where PCC effects should be considered for a correct noise estimate. We evaluated our sensor model for some popular and upcoming UV astronomical instruments and found that there are a few channels in each instrument where our model implies that the SNR is better than the traditional model would suggest. We also have provided a reference implementation of our sensor model in Python, `optika.sensors`, to make this noise model simple to integrate with existing instrument data processing pipelines.

This work does not consider read noise or other noise sources since those will be specific to the particular camera electronics of each instrument, and also cannot be characterized using a VMR. Instead, this work focuses only on the noise intrinsic to the charge-generation process, since this comparatively consistent across different instruments.

We plan to use this work to model the noise for the EUV Snapshot Imaging Spectrograph (ESIS) (**?**), as well as the Full-sun Ultraviolet Rocket Spectrometer (FURST), sounding-rocket-based spectrographs developed by our research group for observing the solar atmosphere.

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NASA grants XXXX, XXXX and XXXX. This article is an example of an executable paper (**?**), all of the simulations, figures, and tables are generated dynamically when this article is created. This helps prevent mistakes and allows this research to be as repeatable as possible. The code to create this document is available at https://github.com/roytsmart/ccd-noise-paper.